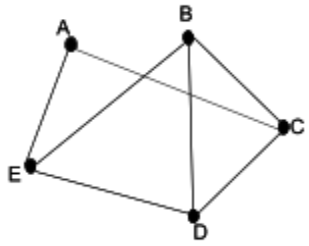
American Computer Science League

2022-2023 ● Contest 4: Short Problems ● Elementary Division

1. Graph Theory

How many edges must be added to the following graph to make it a complete graph?



A. 1
B. 2
C. 3
D. 4
E. 5

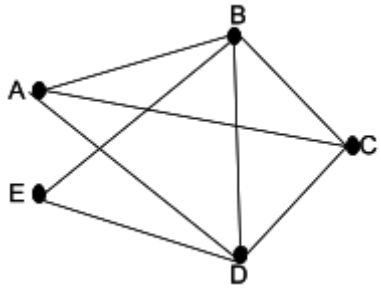
2. Graph Theory

Your job at the ACSL Pizza Shop is to deliver some pizzas to three different locations: a house, a school, and a motel. How many different routes can be used to deliver the pizzas to all 3 locations?

A. 4
B. 6
C. 8
D. 10
E. 24

3. Graph Theory

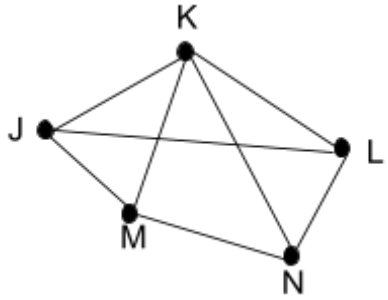
In a rumor network, a message is passed from one person to another in many ways. If this graph represents a rumor network among 5 good friends (Amy, Bob, Carl, Deb, and Evan), how many ways can Amy get a message to Evan using exactly 2 other friends?



A. 1
B. 2
C. 3
D. 4
E. 5

4. Graph Theory

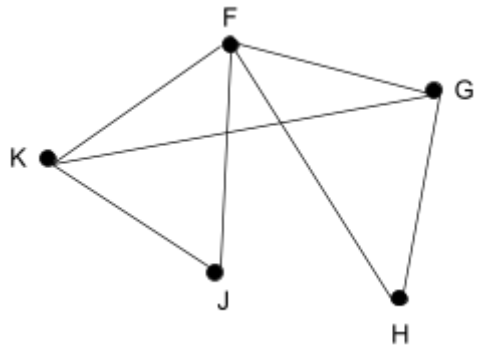
How many cycles of length 3 are in the following graph?



A. 8
B. 6
C. 4
D. 2
E. 0

5. Graph Theory

Which edge could be added to make the following graph traversable from any vertex to any other vertex?



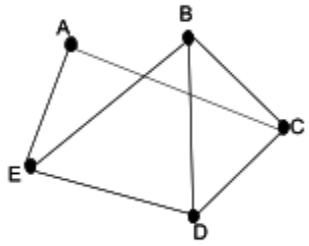
A. HJ
B. GJ
C. HK
D. More than 1 of them
E. It can't be done

American Computer Science League

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1. 图论

以下图形必须添加多少条边，**才能使得它**成为一个**完全图**？



A. 1
B. 2
C. 3
D. 4
E. 5

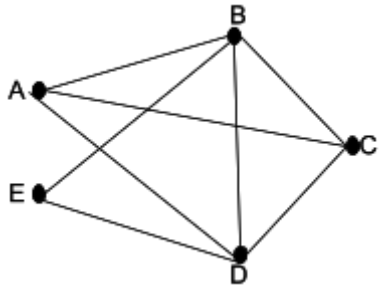
2. 图论

你在 ACSL 披萨店的工作是把一些披萨送到三个不同的地方：一**个住所**、一所学校和一家汽车旅馆。**你**可以使用多少条不同的路线将披萨送到这三个地方？

A. 4
B. 6
C. 8
D. 10
E. 24

3. 图论

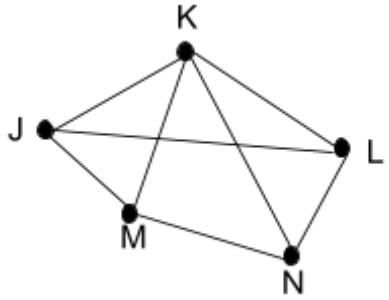
在谣言网络中，信息通过多种方式从一个人传递给另一个人。如果这张图代表了 5 个好朋友（Amy、Bob、Carl、Deb 和 Evan）之间的谣言网络，那么Amy可以通过多少种方式**恰好通过**另外两个朋友向 Evan 传递消息？



A. 1
B. 2
C. 3
D. 4
E. 5

4. 图论

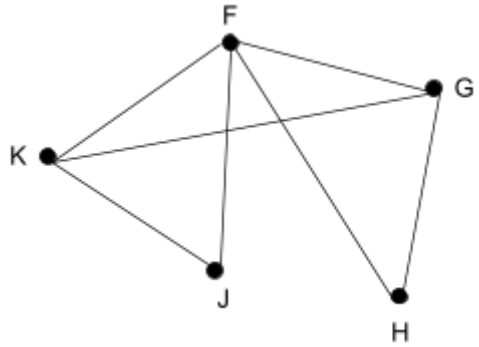
下图中有多少个长度为 3 的环？



A. 8
B. 6
C. 4
D. 2
E. 0

5. 图论

添加哪条边**可以使得**以下图形可以从任何顶点遍历到其他任何顶点？



A. HJ
B. GJ
C. HK
D. 不止一条**边**
E. 无法**实现**